System Simulation Report

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File: TFE.homer

Author: Christian ZIBERA

Location: M8W6+M6R, Kinshasa, Democratic Republic of the Congo (4°18,2'S, 15°18,6'E)

Total Net Present Cost: 33 992,96 $

Levelized Cost of Energy ($/kWh): 0,164 $

Notes: Système Hybrid

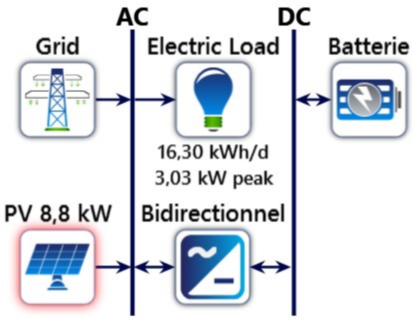
Sensitivity variable values for this simulation

Variable	Value	Unit
Electric Load Scaled Average	15,0	kWh/d

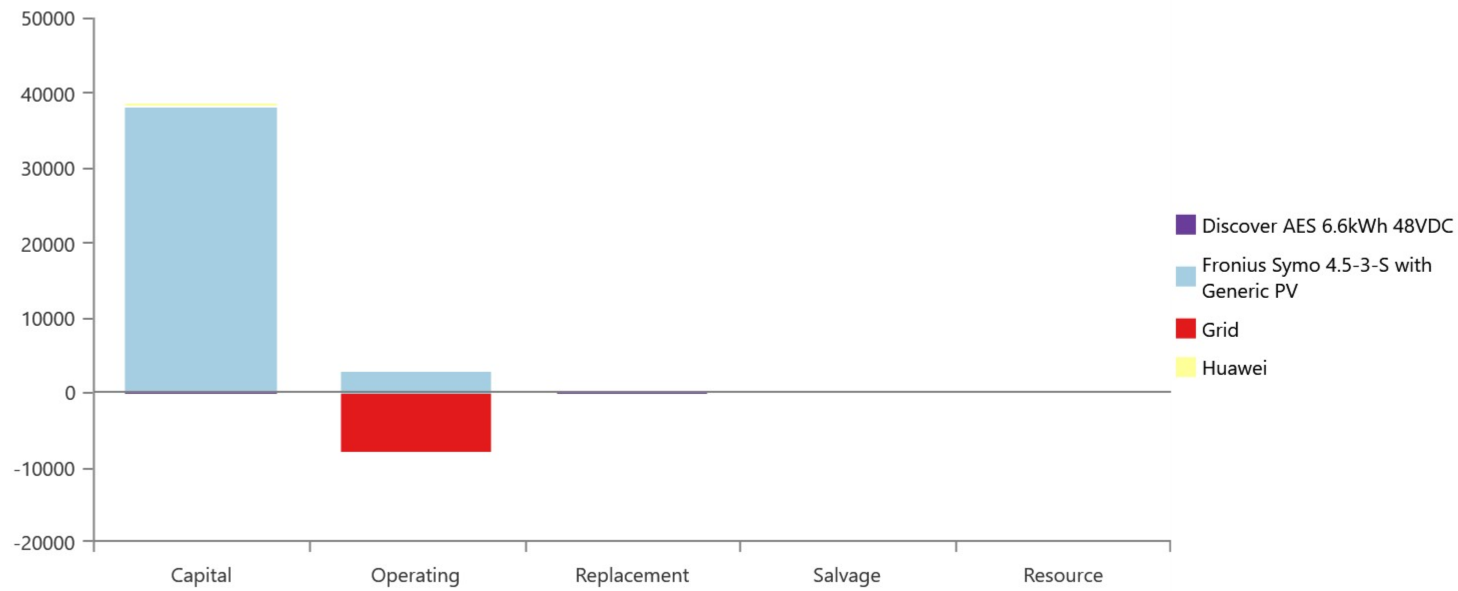
System Architecture

Component	Name	Size	Unit
PV	Fronius Symo 4.5-3-S with Generic PV	8,80	kW
Storage	Discover AES 6.6kWh 48VDC	1	strings
System converter	Huawei	0,104	kW
Grid	Grid	100	kW
Dispatch strategy	HOMER Load Following		

Schematic



Cost Summary



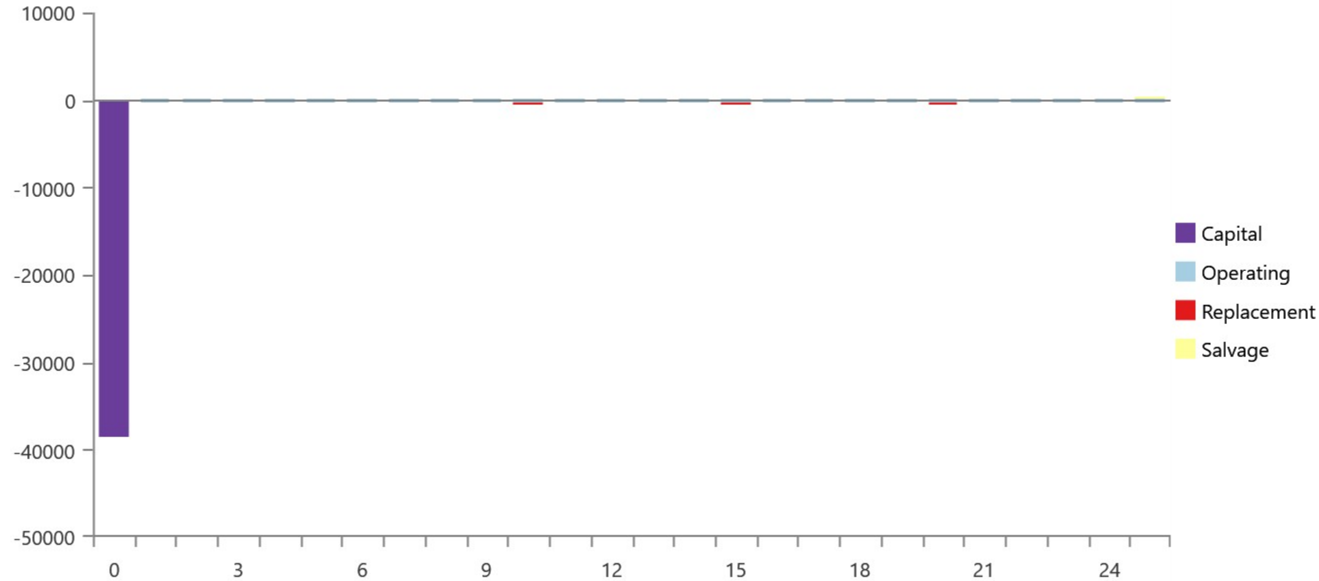
Net Present Costs

Name	Capital	Operating	Replacement	Salvage	Resource	Total
Discover AES 6.6kWh 48VDC	352,00 $	25,86 $	310,97 $	-42,16 $	0,00 $	646,66 $
Fronius Symo 4.5-3-S with Generic PV	38 016 $	2 844 $	0,00 $	0,00 $	0,00 $	40 860 $
Grid	0,00 $	-7 793 $	0,00 $	0,00 $	0,00 $	-7 793 $
Huawei	197,92 $	13,47 $	83,97 $	-15,80 $	0,00 $	279,55 $
System	38 566 $	-4 910 $	394,94 $	-57,97 $	0,00 $	33 993 $

Annualized Costs

Name	Capital	Operating	Replacement	Salvage	Resource	Total
Discover AES 6.6kWh 48VDC	27,23 $	2,00 $	24,05 $	-3,26 $	0,00 $	50,02 $
Fronius Symo 4.5-3-S with Generic PV	2 941 $	220,00 $	0,00 $	0,00 $	0,00 $	3 161 $
Grid	0,00 $	-602,85 $	0,00 $	0,00 $	0,00 $	-602,85 $
Huawei	15,31 $	1,04 $	6,50 $	-1,22 $	0,00 $	21,62 $
System	2 983 $	-379,80 $	30,55 $	-4,48 $	0,00 $	2 630 $

Cash Flow



10000
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12
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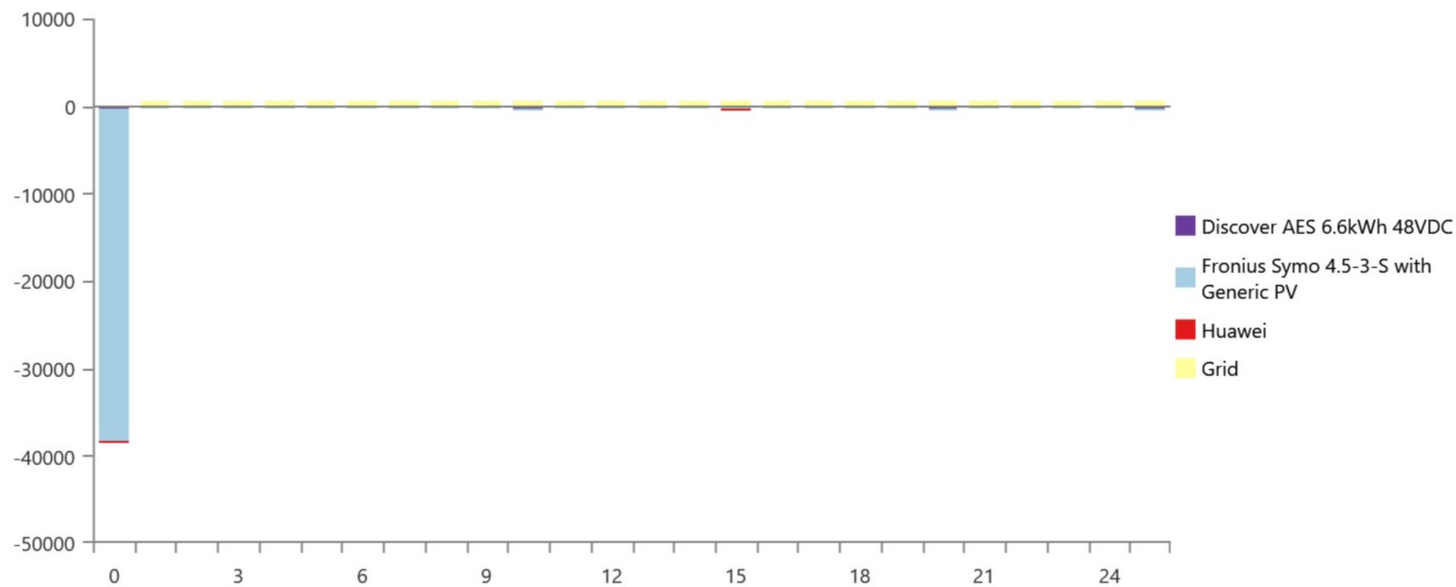
21
24
Capital
Operating
Replacement
Salvage



10000
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Discover AES 6.6kWh 48VDC
Fronius Symo 4.5-3-S with Generic PV
Huawei
Grid

Electrical Summary

Excess and Unmet

Quantity	Value	Units
Excess Electricity	0,0719	kWh/yr
Unmet Electric Load	0	kWh/yr
Capacity Shortage	0	kWh/yr

Production Summary

Component	Production (kWh/yr)	Percent
Fronius Symo 4.5-3-S with Generic PV	13 323	82,8
Grid Purchases	2 759	17,2
Total	16 082	100

Consumption Summary

Component	Consumption (kWh/yr)	Percent
AC Primary Load	5 475	34,1
DC Primary Load	0	0
Grid Sales	10 588	65,9
Total	16 063	100

PV: Fronius Symo 4.5-3-S with Generic PV

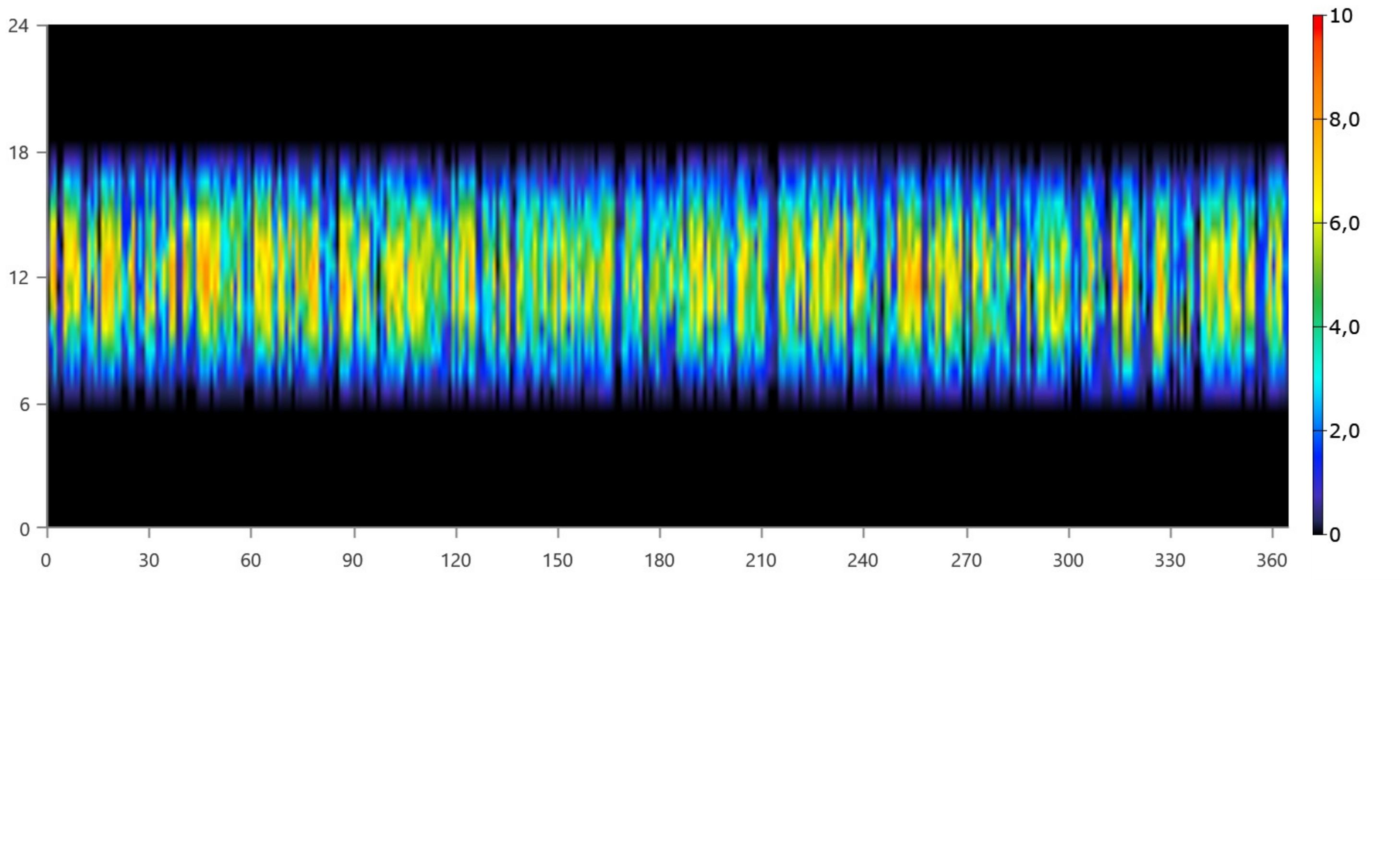
Fronius Symo 4.5-3-S with Generic PV Electrical Summary

Quantity	Value	Units
Minimum Output	0	kW
Maximum Output	8,73	kW
PV Penetration	243	%
Hours of Operation	4 380	hrs/yr
Levelized Cost	0,237	$/kWh

Fronius Symo 4.5-3-S with Generic PV Statistics

Quantity	Value	Units
Rated Capacity	8,80	kW
Mean Output	1,52	kW
Mean Output	36,5	kWh/d
Capacity Factor	17,3	%
Total Production	13 323	kWh/yr

Fronius Symo 4.5-3-S with Generic PV Output (kW)



Storage: Discover AES 6.6kWh 48VDC

Discover AES 6.6kWh 48VDC Properties

Quantity	Value	Units
Batteries	1,00	qty.
String Size	1,00	batteries
Strings in Parallel	1,00	strings
Bus Voltage	48,0	V

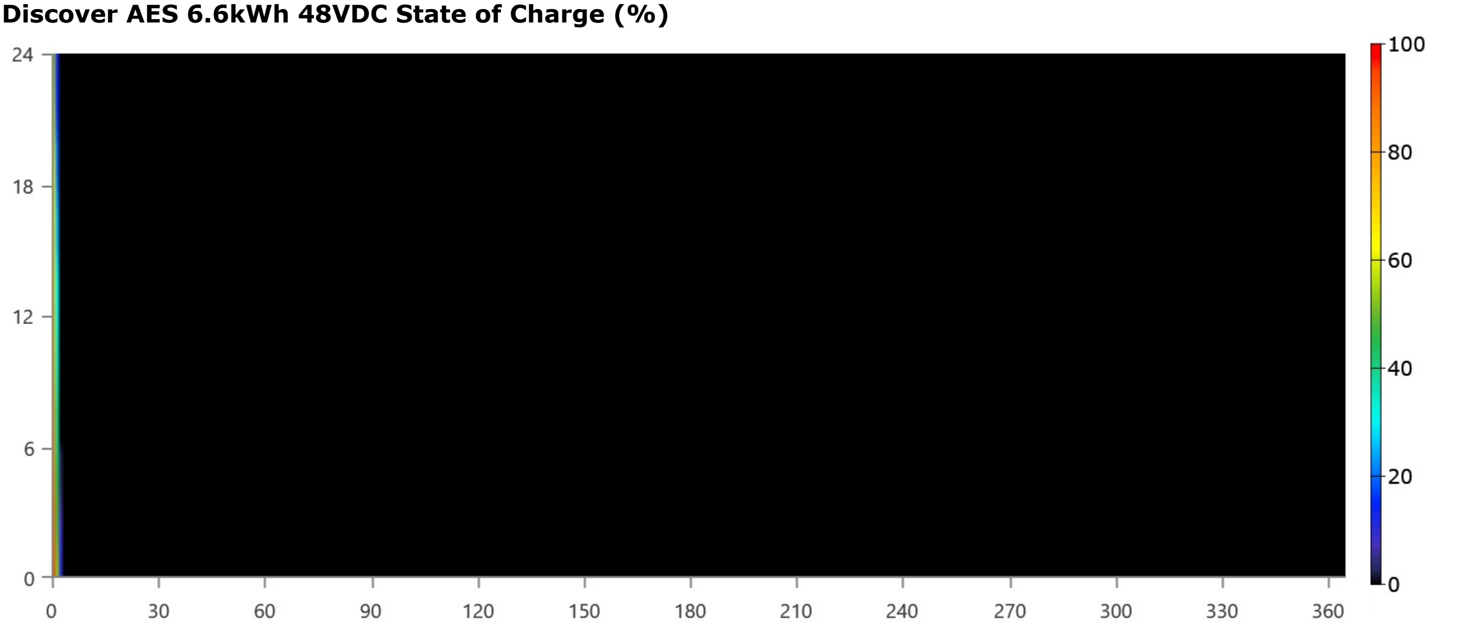
Discover AES 6.6kWh 48VDC Result Data

Quantity	Value	Units
Average Energy Cost	0	$/kWh
Energy In	192	kWh/yr
Energy Out	188	kWh/yr
Storage Depletion	6,24	kWh/yr
Losses	9,74	kWh/yr
Annual Throughput	193	kWh/yr

Discover AES 6.6kWh 48VDC Statistics

Quantity	Value	Units
Autonomy	9,98	hr
Storage Wear Cost	0,00950	$/kWh
Nominal Capacity	6,24	kWh
Usable Nominal Capacity	6,24	kWh
Lifetime Throughput	1 930	kWh
Expected Life	10,0	yr

Discover AES 6.6kWh 48VDC State of Charge (%)



Converter: Huawei

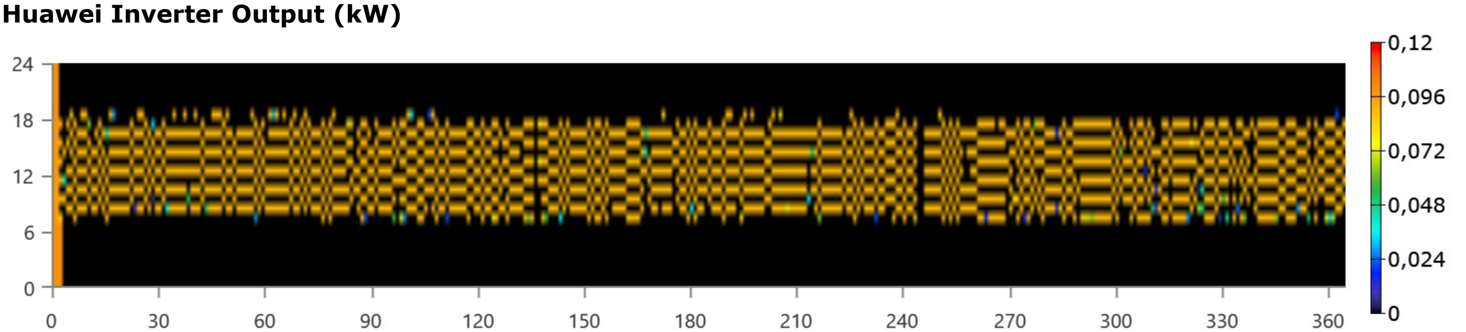
Huawei Electrical Summary

Quantity	Value	Units
Hours of Operation	2 113	hrs/yr
Energy Out	180	kWh/yr
Energy In	188	kWh/yr
Losses	7,52	kWh/yr

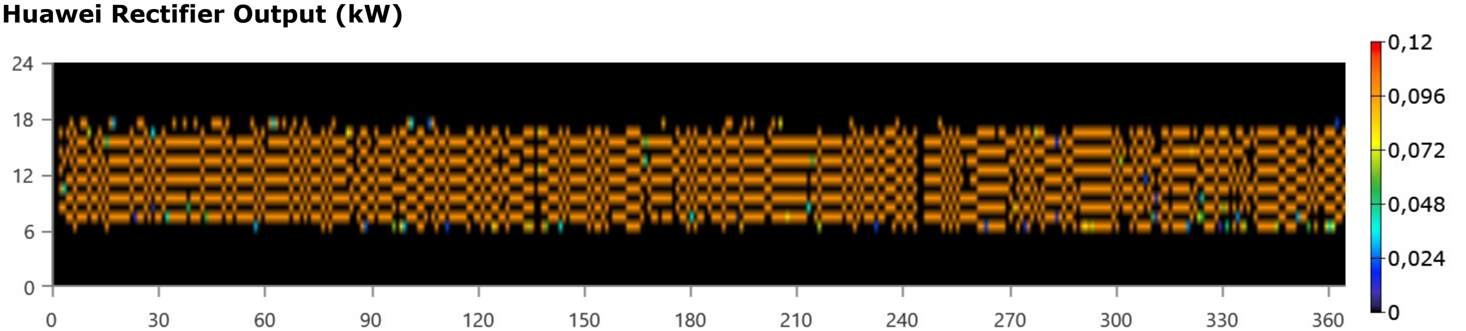
Huawei Statistics

Quantity	Value	Units
Capacity	0,104	kW
Mean Output	0,0206	kW
Minimum Output	0	kW
Maximum Output	0,104	kW
Capacity Factor	19,8	%

Huawei Inverter Output (kW)



Huawei Rectifier Output (kW)



Grid: Grid

Grid rate: Demand 1

Month	Energy Purchased (kWh)	Energy Sold (kWh)	Net Energy Purchased (kWh)	Peak Demand (kW)	Energy Charge	Demand Charge
January	0	0	0	2,57	0,00 $	0,00 $
February	0	0	0	2,25	0,00 $	0,00 $
March	0	0	0	2,53	0,00 $	0,00 $
April	0	0	0	2,58	0,00 $	0,00 $
May	0	0	0	2,33	0,00 $	0,00 $
June	0	0	0	2,17	0,00 $	0,00 $
July	0	0	0	2,59	0,00 $	0,00 $
August	0	0	0	2,48	0,00 $	0,00 $
September	0	0	0	2,49	0,00 $	0,00 $
October	0	0	0	2,38	0,00 $	0,00 $
November	0	0	0	2,79	0,00 $	0,00 $
December	0	0	0	2,32	0,00 $	0,00 $
Annual	0	0	0	2,79	0,00 $	0,00 $

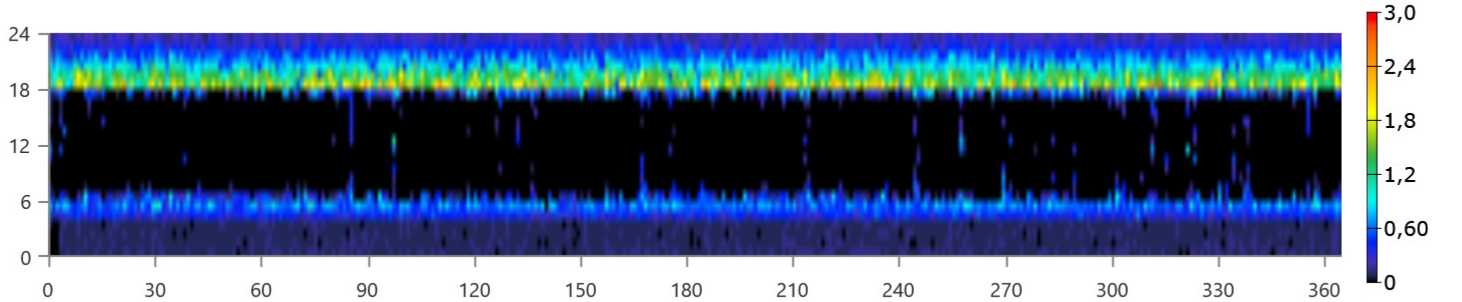
Grid rate: Rate 1

Month	Energy Purchased (kWh)	Energy Sold (kWh)	Net Energy Purchased (kWh)	Peak Demand (kW)	Energy Charge	Demand Charge
January	231	908	-678	0	-52,18 $	0,00 $
February	204	878	-674	0	-51,88 $	0,00 $
March	236	1 001	-765	0	-58,90 $	0,00 $
April	228	942	-714	0	-54,95 $	0,00 $
May	230	889	-659	0	-50,73 $	0,00 $
June	230	827	-597	0	-45,96 $	0,00 $
July	231	925	-693	0	-53,40 $	0,00 $
August	238	941	-703	0	-54,16 $	0,00 $
September	232	874	-642	0	-49,40 $	0,00 $
October	231	787	-556	0	-42,79 $	0,00 $
November	233	785	-551	0	-42,46 $	0,00 $
December	234	832	-598	0	-46,04 $	0,00 $
Annual	2 759	10 588	-7 829	0	-602,85 $	0,00 $

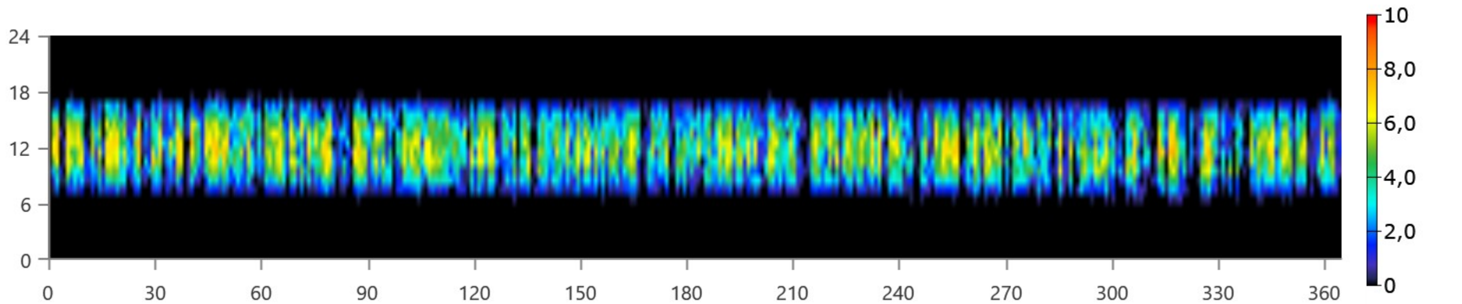
Grid rate: All

Month	Energy Purchased (kWh)	Energy Sold (kWh)	Net Energy Purchased (kWh)	Peak Demand (kW)	Energy Charge	Demand Charge
January	231	908	-678	2,57	-52,18 $	0,00 $
February	204	878	-674	2,25	-51,88 $	0,00 $
March	236	1 001	-765	2,53	-58,90 $	0,00 $
April	228	942	-714	2,58	-54,95 $	0,00 $
May	230	889	-659	2,33	-50,73 $	0,00 $
June	230	827	-597	2,17	-45,96 $	0,00 $
July	231	925	-693	2,59	-53,40 $	0,00 $
August	238	941	-703	2,48	-54,16 $	0,00 $
September	232	874	-642	2,49	-49,40 $	0,00 $
October	231	787	-556	2,38	-42,79 $	0,00 $
November	233	785	-551	2,79	-42,46 $	0,00 $
December	234	832	-598	2,32	-46,04 $	0,00 $
Annual	2 759	10 588	-7 829	2,79	-602,85 $	0,00 $

Energy Purchased From Grid (kW)



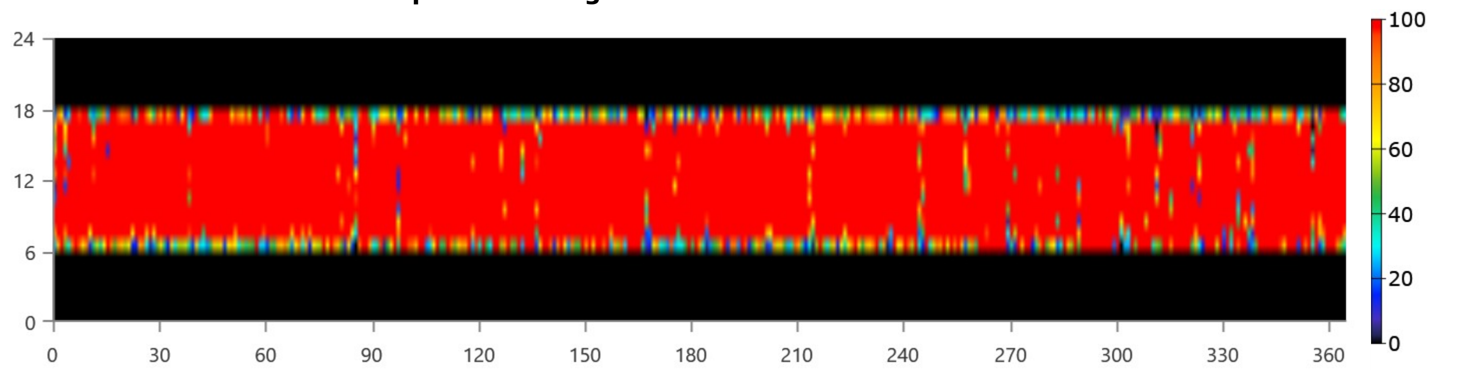
Energy Sold To Grid (kW)



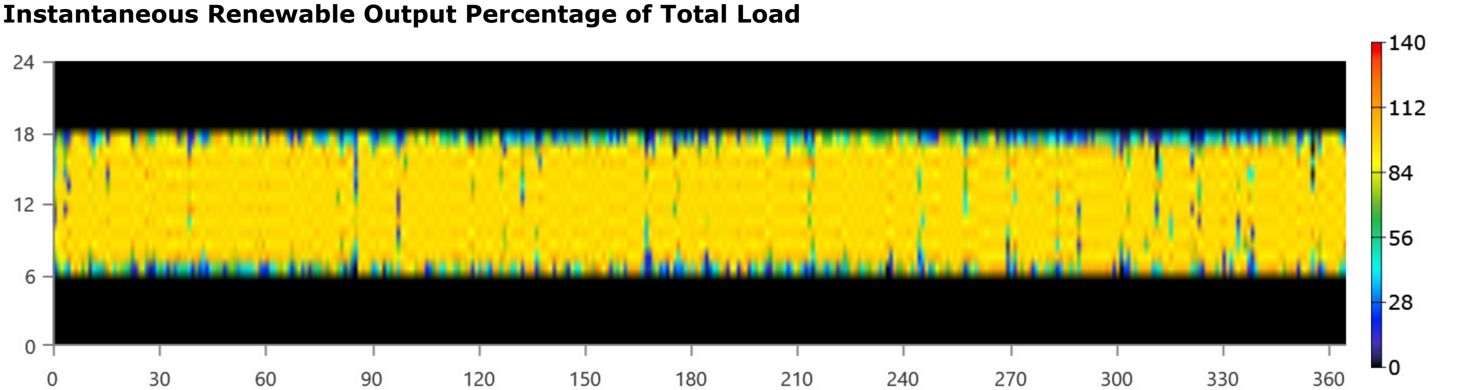
Renewable Summary

Capacity-based metrics	Value	Unit
Nominal renewable capacity divided by total nominal capacity	100	%
Usable renewable capacity divided by total capacity	100	%
Energy-based metrics	**Value**	**Unit**
Total renewable production divided by load	82,9	%
Total renewable production divided by generation	82,8	%
One minus total nonrenewable production divided by load	100	%
Peak values	**Value**	**Unit**
Renewable output divided by load (HOMER standard)	127	%
Renewable output divided by total generation	100	%
One minus nonrenewable output divided by total load	100	%

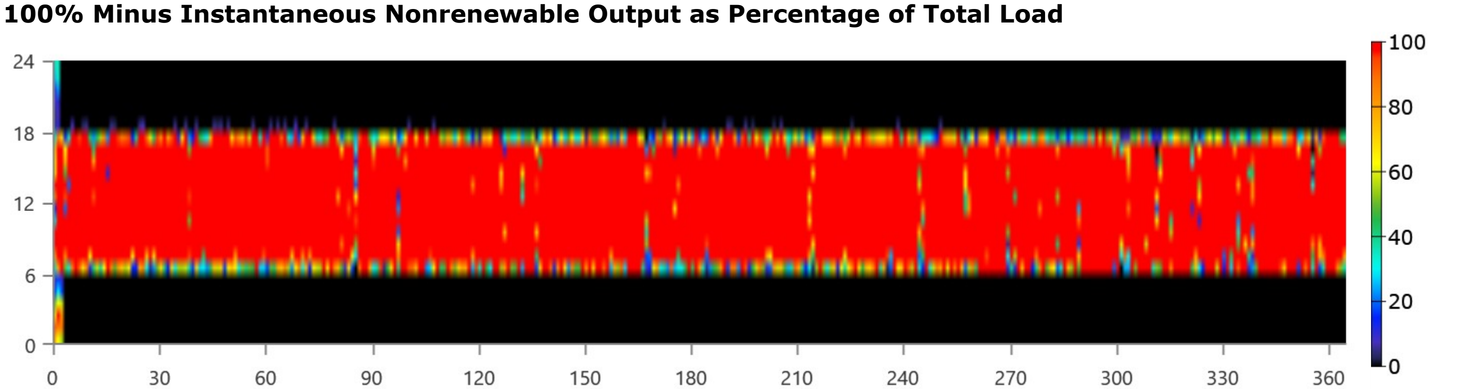
Instantaneous Renewable Output Percentage of Total Generation



Instantaneous Renewable Output Percentage of Total Load



100% Minus Instantaneous Nonrenewable Output as Percentage of Total Load



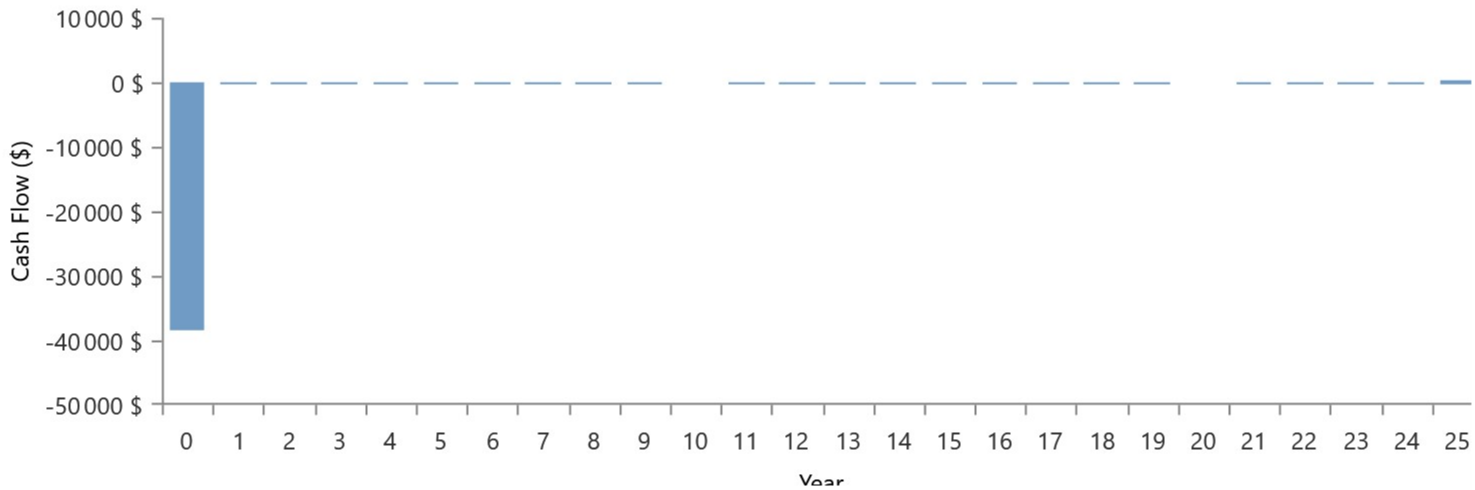
Compare Economics

Architecture									Cost	
					PV 8,8 kW (kW)	Batterie	Grid (kW)	Bidirectionnel (kW)	NPC ($)	Initial capital ($)
							100		5 450 $	0,00 $
					8,80	1	100	0,104	33 993 $	38 566 $

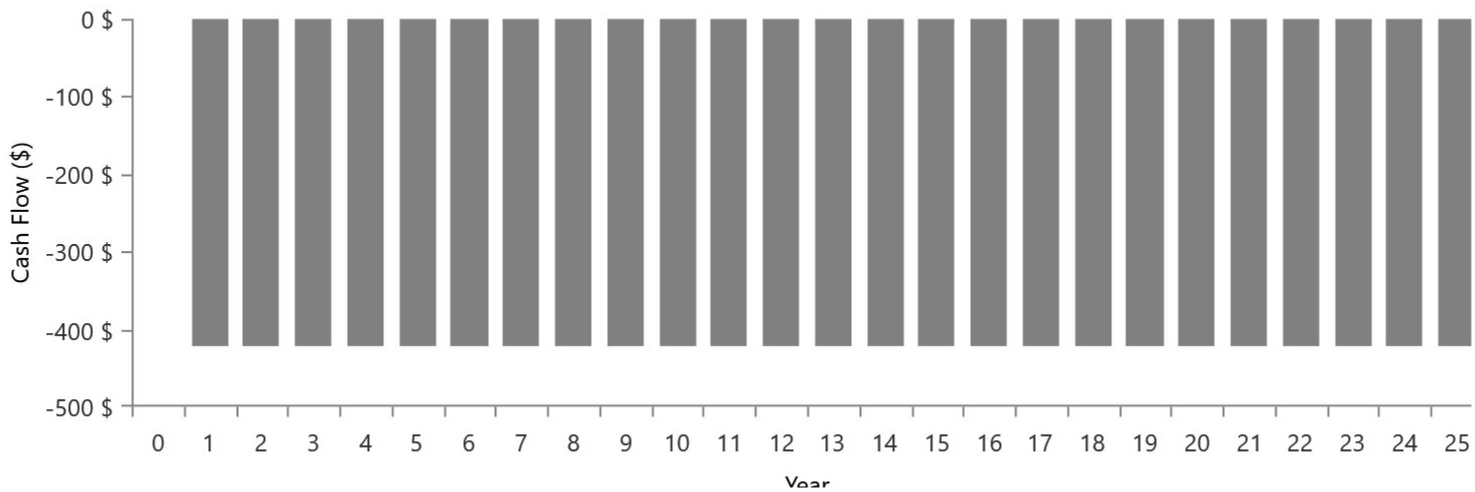
Base case is highlighted in green.

Metric	Value
Present worth ($)	-28 543 $
Annual worth ($/yr)	-2 208 $
Return on investment (%)	-2,0
Internal rate of return (%)	n/a
Simple payback (yr)	n/a
Discounted payback (yr)	n/a

Current Annual Nominal Cash Flows



Base Case Annual Nominal Cash Flows



Cumulative Discounted Cash Flows